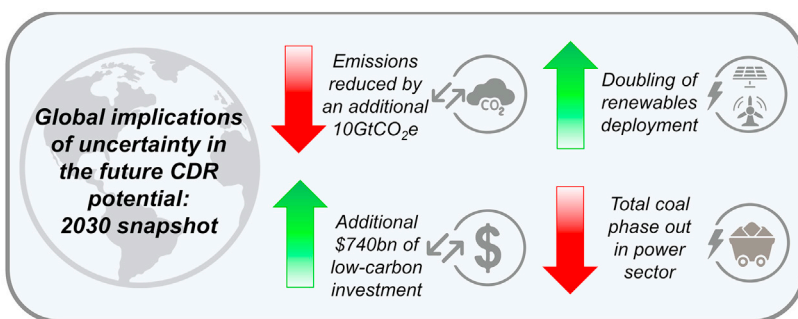
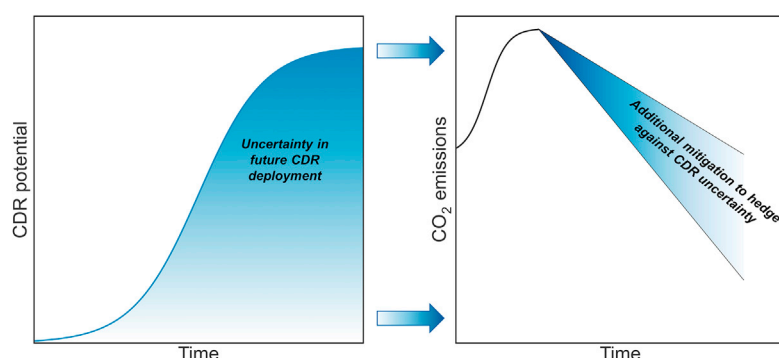


Report

The policy implications of an uncertain carbon dioxide removal potential

Uncertainty in the future CDR potential leads to greater near-term action to reduce emissions



We conduct an expert survey into the feasible level of carbon dioxide removal (CDR). Experts suggest that feasible CDR deployment could be large but remains highly uncertain. We explore the implications of this uncertainty for the critical decade of the 2020s. Uncertainty in the future CDR potential should encourage greater climate action in the 2020s, with deeper emissions cuts, accelerated renewables deployment, greater investment in low-carbon technology, and a faster phaseout of fossil fuels from the energy system.

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Highlights

Experts are surveyed on the feasible CDR potential from BECCS, DACCS, and afforestation

Results highlight a potentially large but highly uncertain CDR potential

Uncertainty in future CDR potential drives greater climate action in the 2020s

High uncertainty in CDR leads to an extra 10 GtCO₂e of emissions reductions by 2030

Report

The policy implications of an uncertain carbon dioxide removal potential

Neil Grant,^{1,3,*} Adam Hawkes,² Shivika Mittal,¹ and Ajay Gambhir¹

SUMMARY

Many low-carbon scenarios rely on carbon dioxide removal (CDR) to meet decarbonization goals. The feasibility of large-scale CDR deployment is highly uncertain, and existing scenarios have been criticized for overreliance on CDR. We conduct an expert survey on the feasible potential for CDR via bioenergy with carbon capture and storage, direct air capture and afforestation. We use the survey results to represent uncertainty in future CDR availability and explore the implications in an integrated assessment model. Stochastic optimization demonstrates that uncertainty in future CDR availability provides a strong rationale to increase near-term rates of decarbonization. In scenarios with high CDR uncertainty, emissions are reduced by an additional 10 GtCO₂e in 2030 compared with scenarios with no consideration of CDR uncertainty. This highlights the urgent need to increase ambition contained in nationally determined contributions (NDCs) for 2030, to get the world on track to deliver 1.5°C and to hedge against an uncertain future CDR potential.

INTRODUCTION

Carbon dioxide removal (CDR) is the intentional removal of CO₂ from the atmosphere, through engineered methods and the enhancement of natural carbon sinks.¹ CDR features heavily in existing low-carbon scenarios produced by integrated assessment models (IAMs).² In the IPCC's Special Report on 1.5°C, scenarios that limit warming to 1.5°C with no or limited overshoot remove 6.5 GtCO₂/year by 2050, and over the century 670 GtCO₂ is removed from the atmosphere via CDR³ (median projections), an amount greater than the remaining carbon budget for 1.5°C² and which represents over 16 years of current CO₂ emissions.⁴

While studies have highlighted the value of CDR in tackling climate change,^{5–9} the feasibility of large-scale deployment remains highly uncertain,¹⁰ and there are widespread concerns around the risk of relying on CDR to meet climate goals.^{11–18} These include the physical risks of temporarily exceeding 1.5°C via overshoot scenarios,¹¹ the potential for mitigation deterrence,^{12–14} the financial viability of large-scale CDR deployment,^{15,16} the side effects of certain CDR technologies,¹⁷ and the possibility that CDR failure could then jeopardize the Paris Agreement¹⁸ with severe consequences for future generations and especially the global poor. It is essential that low-carbon scenarios informing policymakers incorporate CDR in an appropriate manner, cognizant of its risks and ethical concerns.¹⁹

We make three contributions to long-standing debates on the appropriate representation of CDR in scenario analysis. First, we present the results of an expert survey

Context & scale

Carbon dioxide removal (CDR) may be a valuable tool for tackling the climate crisis but remains controversial, due to the risk that anticipated future CO₂ removal could reduce incentives to cut emissions now. Low-carbon scenarios that inform decision making should limit CDR deployment to feasible levels and explore the implications of uncertainty in future CDR deployment.

We conduct the first expert survey on the feasible scale of CDR deployment and find that large-scale CO₂ removal may be possible, but its feasibility remains highly uncertain. This uncertainty provides a strong rationale to accelerate climate action in the present and avoid the risks of betting on CDR, which could then fail to materialize. Accounting for uncertainty in future, CDR leads to an additional 10 GtCO₂e of emission reductions by 2030, driven by greater investment in renewables. This provides further justification to increase the pace of climate action and truly make the 2020s the decade of transformation.



on the feasible potential for achieving CDR via three methods commonly represented in IAMs: bioenergy with carbon capture and storage²⁰ (BECCS), direct-air carbon capture and storage²¹ (DACCS), and afforestation/reforestation (AR).²² Second, we use survey results to conduct an informed evaluation of existing scenarios produced by IAMs. To date, IAM scenarios appear to substantially overestimate the potential for CDR via BECCS, while there is greater agreement between experts and IAMs in the case of DACCS and AR. Finally, we incorporate the survey results into an integrated assessment model, using stochastic optimization to explore the policy implications of uncertainty in CDR availability. Greater consideration of uncertainty in CDR availability leads to faster emission reductions driven predominantly by the deployment of renewables. These results provide further justification to ratchet up the strength of climate targets for 2030 and truly make the 2020s the decade of transformation.

RESULTS

Experts' view on the potential for CDR

We conduct an expert survey exploring the feasible potential for CDR. The feasible potential was presented to experts as a subset of the technical potential for CDR, accounting for non-technical constraints such as sustainability considerations and societal/governance issues such as institutional requirements ([experimental procedures](#) and [Notes S1](#) and [S2](#)).

[Figure 1](#) displays the survey results for each CDR method considered in this analysis. Expert opinion on the feasible potential for CDR varies strongly across methods. Experts judge AR to have the largest potential pre-2050. However, in the long-term AR has the lowest potential, with experts citing issues around land availability, institutional challenges, and sink saturation. With high costs and low technological readiness, DACCS has limited potential pre-2050, but as costs fall and carbon prices rise, experts suggested that DACCS deployment could grow strongly, with DACCS having the greatest long-term potential. When considering cumulative sequestration across the century, experts judge BECCS to have the lowest feasible potential, with a median estimate of 196 GtCO₂ feasibly removed from 2020 to 2100. DACCS has the largest cumulative potential, with a median feasible potential estimate of 320 GtCO₂. DACCS also demonstrated the greatest degree of variance between experts, with some judging the feasible potential to be in excess of 1,000 GtCO₂ over the century. Across all three methods, the feasible potential for CDR remains highly uncertain, with experts providing varied and contrasting views on feasible levels of CO₂ removal.

[Figure 2](#) compares the results of the expert survey with CDR deployment from a range of low-carbon scenarios produced by IAMs.^{3,7,23–27} BECCS deployment in the 1.5°C-compatible IAM scenarios reviewed substantially exceeds the feasible potential as given by expert opinion. Experts estimate that median cumulative sequestration from BECCS could approach 200 GtCO₂ by 2100, compared with 540 GtCO₂ in reviewed IAM scenarios. Median DACCS deployment in IAMs is greater than the feasible potential derived from the expert survey, but there is better agreement between experts and IAMs in this case, with substantial overlap between the distributions. Of the methods evaluated, there is the greatest agreement with expert opinion in the case of AR, for which cumulative sequestration in reviewed scenarios and expert opinion are closely matched. Similar results can be seen for BECCS/AR when comparing the survey results to 2°C scenarios ([Figure S1](#)). In the case of DACCS, substantially reduced deployment in 2°C pathways means that reviewed

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<https://doi.org/10.1016/j.joule.2021.09.004>

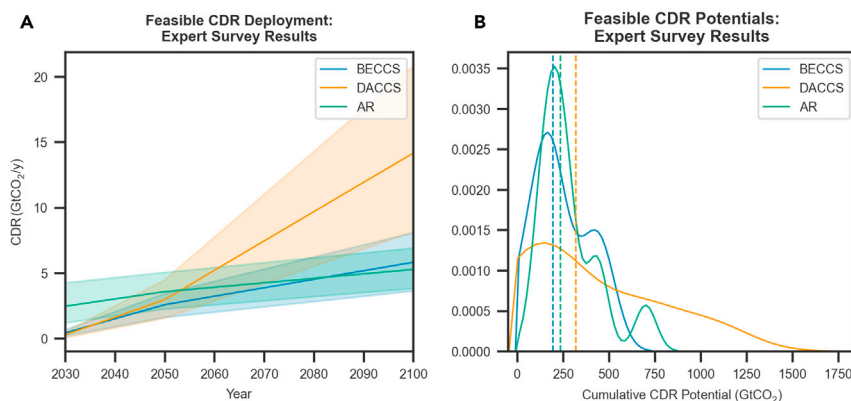


Figure 1. Expert opinion on the feasible potential for CDR

Figure displays the survey results for all three CDR methods assessed.

(A) Mean feasible potential for CDR as a function of time, with a 95% confidence interval in the mean calculated via bootstrapping. We use $n_{boot} = 1,000$ (Note S2).

(B) Estimated probability density functions for cumulative CDR from 2020 to 2100 for each method, with the median for each CDR route given by the dashed line. These density functions are reweighted and normalized to ensure that the function is only defined for cumulative sequestration >0 (Note S2). The mean CDR potential as estimated by experts is given in Table S3.

IAM scenarios sequester less CO₂ via DACCS than the feasible potential as given by expert opinion (Note S3).

Transformative action: the policy implications of an uncertain potential for CDR

Actions taken to reduce emissions today occur under deep uncertainty around the feasible scale of future CDR. Most low-carbon scenarios are, however, deterministic—with all variables known with perfect certainty over the whole time horizon. This does not represent the real-world decision-making context of policymakers appropriately.¹² It is essential to understand how uncertainty in the availability of future technologies should affect near-term climate policy.^{12,28}

The survey results highlight considerable uncertainty in feasible CDR potential, with a wide range of potentials suggested by experts. Using a detailed-process IAM (TIAM-Grantham), we use stochastic optimization to explore the implications of this uncertainty in CDR for climate policy in the 2020s. We analyze scenarios that limit end-of-century warming to 1.5°C and explore four different levels of uncertainty in future CDR availability by representing an increasing variety of possible future CDR potentials (experimental procedures). We use the expert survey results to represent the probability distribution of CDR availability. The use of survey data is intended to be illustrative, and similar insights could be generated with other representations of CDR uncertainty, including *ad hoc* or stylized distributions.

Representing uncertainty in future CDR leads to TIAM-Grantham taking greater action to reduce emissions in the 2020s (Figure 3). Moving from left to right in Figure 3 corresponds to increased uncertainty around the availability of CDR—from deterministic scenarios in which future CDR availability is known with certainty, to scenarios with high uncertainty around the feasible CDR potential (which could therefore be very large or very small). Increased representation of uncertainty in CDR leads to scenarios with faster emission reductions, higher carbon prices, greater renewables deployment, and a faster phaseout of fossil fuels across the energy sector. This suggests that, given uncertainty in future CDR deployment, it is

CDR deployment in IAMs and via expert views

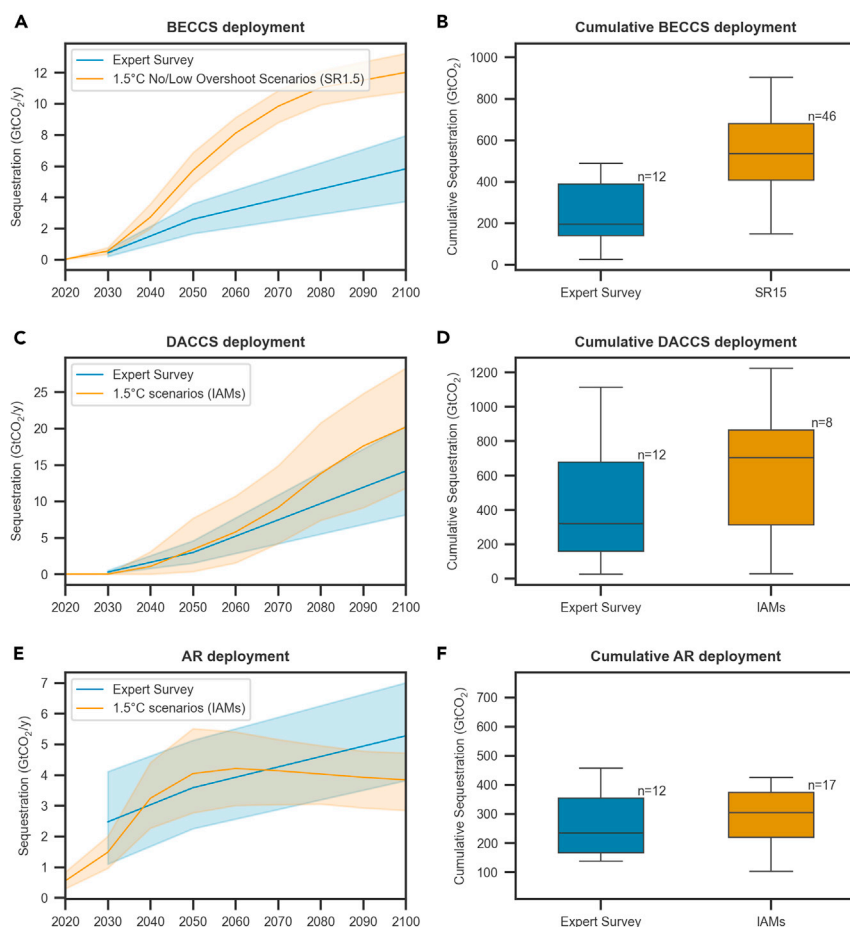


Figure 2. Comparing CDR deployment in IAMs to the feasible potential as given by the expert survey

Figure compares CDR deployment from selected 1.5°C scenarios produced by IAMs with the results of the expert survey. (A, C, and E) Mean CDR deployment as a function of time, with a 95% confidence interval in the mean calculated via bootstrapping. We use $n_{boot} = 1,000$ (Note S3). The potential is given by sequestration rates (GtCO₂/year) and accounts for plant decommissioning over time.

(B, D, and F) Cumulative sequestration over the 2020–2100 period, with boxes displaying the median and interquartile range.

Data were taken from SR1.5, with AR/DACCS supplemented by studies published post-SR1.5 to increase the limited dataset. The number of experts/scenarios contributing to each CDR method is given by *n* in the right-hand panels.

suboptimal climate policy to relax the pace of emission reductions in the 2020s and hope to compensate for this later by large-scale negative emissions. Instead, faster near-term decarbonization emerges as a low-regrets option to insure against the possibility of limited future removals.

In the stochastic scenarios analyzed, greater uncertainty in CDR availability results in deeper decarbonization in the 2020s. This occurs because increased representation of CDR uncertainty better captures the tails of the probability distribution, where CDR potential is very high or very low. It is less costly to underestimate the future CDR potential (and mitigate faster than strictly necessary in the near term) than to

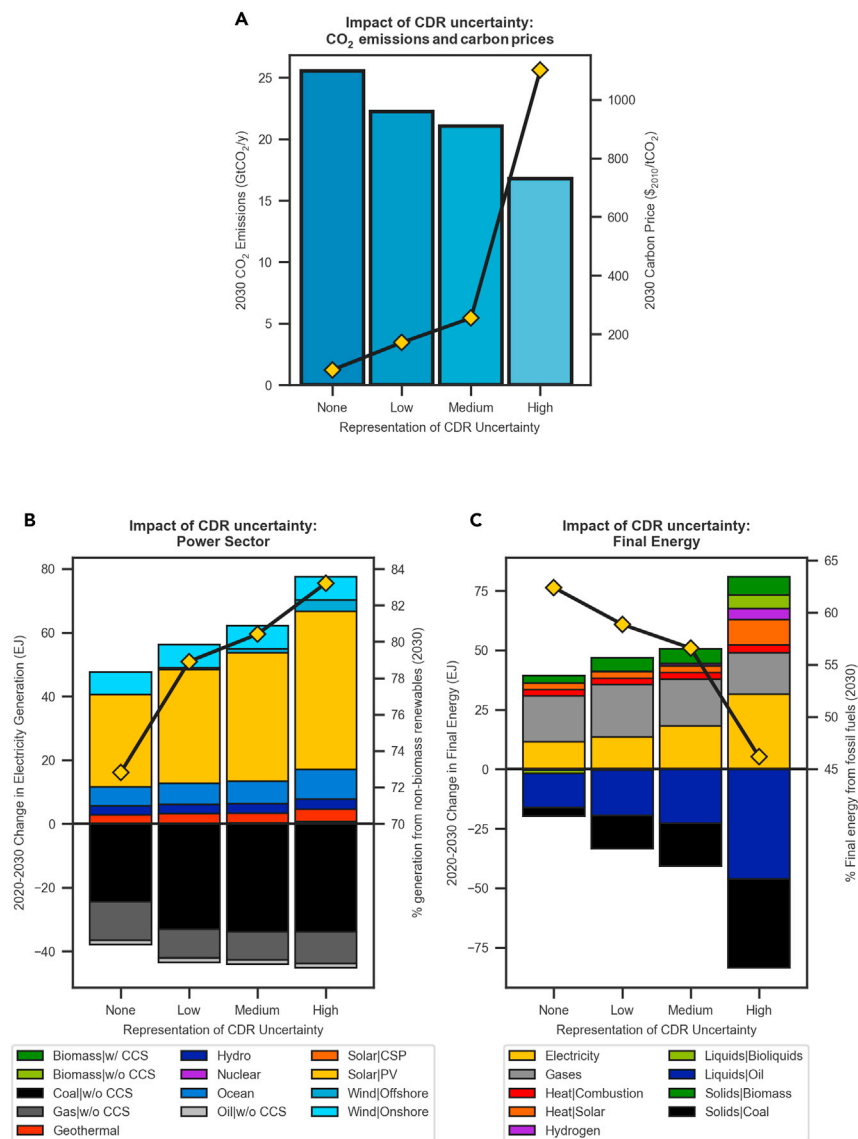


Figure 3. Transformative action: the impact of uncertainty in CDR on decision making for the 2020s

Figure displays how increased representation of uncertainty around CDR availability, via greater sampling of the expert-informed probability distribution for CDR, affects near-term climate policy. (A) Anthropogenic CO₂ emissions and carbon prices in 2030.

(B) Change in electricity generation from 2020 to 2030 and share of non-biomass renewables in electricity generation.

(C) Change in final energy from 2020 to 2030 and share of final energy from fossil fuels.

In all plots, left-hand axes refer to the bar charts, and right-hand axes refer to the line plots. No/Low/Medium/High representation of CDR uncertainty corresponds to sampling the expert-informed CDR probability distribution one/two/three/four times, and thus considering an increasing range of possible future CDR potentials when performing stochastic optimization (Table 2 and experimental procedures).

overestimate the future CDR potential (undertaking insufficient near-term mitigation and betting on CDR which then fails to materialize, with associated large transitional costs). This means that, for any given level of CDR, the cost function is asymmetric around the optimum level of emissions (Figure 4).

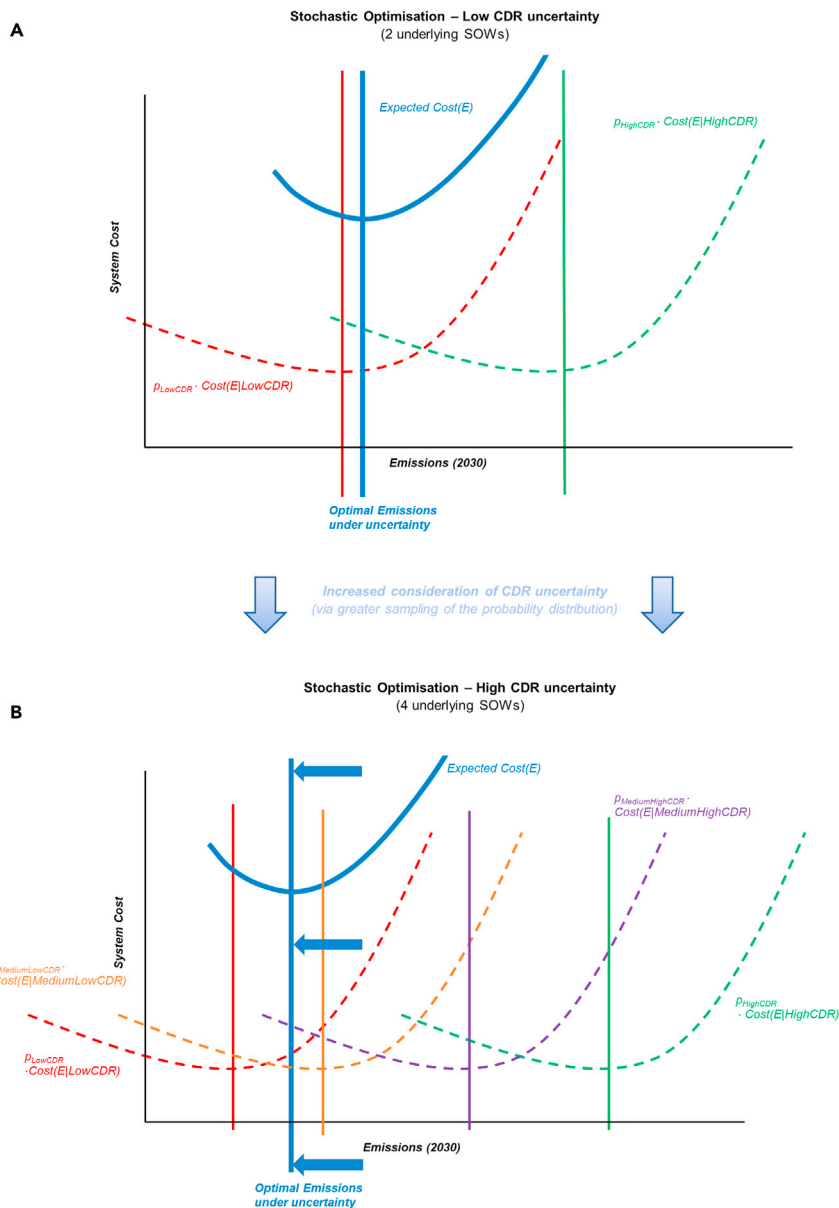


Figure 4. Expected cost as a function of 2030 emissions

Figure illustrates the expected cost function (blue), as a function of emissions in 2030.

(A) Low CDR uncertainty: two future states-of-world (SOWs) considered. A high-CDR world in which near-term emissions can remain relatively high (green) and a low-CDR world in which near-term emissions must fall rapidly (red) are represented.

(B) High CDR uncertainty: four future SOWs considered. High (green), low (red), medium-low (orange), and medium-high (purple) CDR potentials are represented.

Vertical lines show the optimal emissions for a given CDR potential. The cost functions are asymmetric, with the cost of betting on CDR which fails to materialize (higher near-term emissions) much greater than the cost of underestimating the future potential for CDR (lower near-term emissions). As a result, the expected cost function's minima are weighted heavily toward the possibility of low-CDR outcomes, and optimal emissions under uncertainty are at the lower end of the range. Increased sampling leads to lower emissions in the optimal hedging strategy, due to this asymmetry (blue arrow in B shows the reduction in optimal emissions under the hedging strategy). Curves are illustrative.

Stochastic optimization considers all underlying possible CDR potentials when calculating the expected cost across all emission levels and thus determining the optimal level of action in the hedging strategy ([experimental procedures](#)). This is given by the minima of the blue curve in [Figure 4](#). Taking the example of low representation of uncertainty (two possible future CDR potentials considered), asymmetry in the underlying cost functions leads to the optimum level of near-term emissions being weighted more strongly toward low-CDR outcomes, rather than high-CDR outcomes ([Figure 4A](#)). When the probability distribution is sampled with greater frequency, the two existing curves move to the left/right as the tails of the distribution are explored more fully, and additional curves (orange and purple) are added to populate the central parts of the distribution. Due to the asymmetry in the underlying cost functions discussed above, increased representation of CDR uncertainty, despite better representing the high-end tail of the distribution as well as the low-end tail, leads to larger and earlier emission reductions ([Figure 4B](#)). In modeled scenarios, it is better to underestimate the contribution that CDR could make to meeting carbon budgets than to overestimate it.

Moving from a scenario with no representation of CDR uncertainty to one with high representation of uncertainty results in anthropogenic CO₂ emissions being reduced an additional 8.8 GtCO₂ by 2030, an amount greater than current CO₂ emissions from the EU and USA combined.⁴ Accounting for all greenhouse gases, high consideration of uncertainty in CDR potential results in an additional 10 GtCO_{2e} of emissions reductions by 2030. This increases the emissions gap between current NDCs and the 1.5°C pathway from 18GtCO_{2e} in 2030 to 28GtCO_{2e}, or by more than 50% ([Note S4](#)).

This near-term decarbonization in TIAM-Grantham is driven by a faster phaseout of coal and oil, with much greater reliance on renewable electricity (particularly solar photovoltaics) and low-carbon fuels (hydrogen, bioenergy, and renewable heat). [Table 1](#) presents some key decarbonization indicators for the 2020s across the four levels of CDR uncertainty considered.

All scenarios see an increase in global energy investments in order to meet the 1.5°C target.²⁹ However, scenarios with higher uncertainty in CDR require greater near-term investment to hedge against the potential of limited future CDR availability. In 2030, energy investments are US\$ 740 billion larger in a scenario with high CDR uncertainty compared to a deterministic scenario with no uncertainty (USD₂₀₁₀). These investments are used to support a rapid expansion of renewable energy to decarbonize the power sector and expand electrification of end-use sectors. In 2020, the world installed 260 GW of new renewable capacity.³⁰ Modeled scenarios deploy between 390 and 750 GW of renewables per year across the 2020s (decadal average), with the rate of deployment tripling in scenarios with high uncertainty around future CDR availability.

A rapid transition away from coal-fired electricity will be essential in meeting the 1.5°C target.³¹ Recent deterministic scenarios have suggested that coal generation must fall by around 70% over the decade.^{2,31} If, however, future CDR availability is considered as uncertain, an even faster coal phaseout is required, with the entire coal-fired power fleet phased out by 2030 in scenarios with medium or high CDR uncertainty. A similar acceleration of the energy system transition is observed in the end-use sectors of transport, buildings, and industry.

Accounting for uncertainty in the feasible potential for CDR is therefore essential to appropriately forming policy decisions for the next decade. The scenarios presented

Table 1. The impact of uncertainty in CDR availability on near-term climate policy

Sector	Metric	No CDR uncertainty	Low CDR uncertainty	Medium CDR uncertainty	High CDR uncertainty
Overview	average emissions reduction rate across 2020s	4.0%/year	5.3%/year	5.8%/year	7.9%/year
Overview	2030 CO ₂ emissions (GtCO ₂ /year)	25.5	22.2	21.1	16.8
Overview	global energy investments over 2020s (relative to 2020)	+25%	+33%	+40%	+57%
Overview	fossil share of final energy in 2030 (%)	62%	59%	57%	46%
Power	average renewables deployment across 2020s (GW/year)	390	500	590	750
Power	reduction in coal-fired generation over 2020–2030 (%)	72%	97%	100%	100%
Power	share of electricity from non-biomass renewables in 2030 (%)	73%	79%	80%	83%
Transport	surface transport: share of electricity/hydrogen in final energy consumption in 2030 (%)	4%	6%	10%	21%
Transport	aviation and shipping: share of sustainable fuels (%)	6%	5%	8%	21%
Transport	reduction in oil consumption over 2020–2030 (%)	17%	22%	26%	48%
Buildings	share of heat from zero-carbon sources in 2030 (%)	25%	29%	34%	41%
Industry	share of industry final energy from electricity/hydrogen/biomass in 2030 (%)	34%	36%	38%	48%

Data from stochastic analysis in TIAM-Grantham, for scenarios that limit end-of-century warming to 1.5°C with a 50% probability.

here are illustrative, and uncertainty would be even better represented by greater sampling of the CDR probability distribution. Further consideration of uncertainty would drive even faster emission reductions, renewables deployment, and fossil-fuel phaseout than presented in these initial scenarios.

DISCUSSION

CDR may well be essential for removing historical and future anthropogenic emissions, if we wish to return toward safe atmospheric concentrations of CO₂.³² However, there remains deep uncertainty around the feasible scale of CDR deployment. We provide an expert-derived estimate of the feasible potential for CDR and use this estimate to demonstrate the implications of uncertainty in future CDR availability for near-term climate policy.

Our results suggest that large-scale CDR may well be feasible, with a median cumulative CDR potential from the expert survey of 740 GtCO₂ by 2100. However, this potential remains highly uncertain, with experts providing varied and contrasting views on feasible levels of CDR. In addition, how carbon removal (at any scale) should feature in climate strategies remains a matter for debate.^{12,33} We have shown that formally representing CDR uncertainty in a detailed-process IAM leads to greater near-term decarbonization, particularly via renewables uptake, as emissions reductions that can be known with certainty should not be substituted for future uncertain removals. Given uncertainty in future CDR availability, policymakers should focus on reducing emissions as fast as possible, while funding RD&D into carbon removal. The best outcome would be the ability to deploy, without having to rely on, CDR at scale.

Reducing reliance on uncertain CDR requires, in modeled scenarios, an additional 10 GtCO₂e of emissions reductions by 2030. This considerable increase in global ambition is not captured by deterministic scenarios, which fail to account for

uncertainty in the feasible CDR potential. Conventional estimates of the emissions gap between current NDCs and Paris-compliant scenarios in 2030 may therefore need to be revised upward, with the benchmark for Paris-compliant NDCs correspondingly higher. The gulf between current NDC targets for 2030 and robust climate policy may be even larger than currently thought. This further emphasizes the need for rapid and transformative action to cut emissions in the 2020s.

Our results further support the implications of new scenario frameworks which limit peak warming,¹¹ demonstrating via a different methodological innovation (optimization under uncertainty) the risks of relying on overshoot and subsequent large-scale CDR deployment to meet the Paris Agreement goals. While this research focuses on CDR, the results have implications for other technological promises with uncertain future potential, demonstrating that uncertain future contributions to decarbonization (whether from CDR or other technologies) should have limited substitutability with near-term emission reductions. Given deep uncertainty around the feasible potential for CDR, scientists, civil society and policymakers should push for maximum climate action in the 2020s, rather than betting on highly uncertain CDR resources.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Neil Grant (n.grant18@imperial.ac.uk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All data presented in this paper are available in the [supplemental information](#) and also at the following <https://doi.org/10.5281/zenodo.5467406>.

Expert survey

The survey asked respondents to provide their best estimate of the feasible CDR potential in 2030, 2050, and 2100 for three different removal options: AR, BECCS, and DACCS. These CDR options were chosen for two reasons. First, as three of the four most studied CDR routes in recently reviewed literature¹ (alongside soil carbon sequestration), they represent options for which there is the greatest available expertise in the academic community. Second, they are the three options most represented in existing IAMs, which are used as a point of comparison with the survey results. The feasible potential for CDR could be greater than suggested by this analysis if other routes for CDR, such as enhanced weathering,³⁴ soil carbon sequestration and biochar,³⁵ or ocean fertilization³⁶ were included. Greater representation of CDR methods in IAMs can lead to more balanced deployment of individual options,²⁶ which could improve agreement between expert opinion and IAM scenarios. Further analysis could extend this research to consider a wider range of CDR options in both expert surveys and IAM scenarios.

The feasible potential was defined as a subset of the technical potential, accounting for non-technical constraints such as social/governance issues and sustainability limits. This is analogous to the use of the term when estimating renewable potentials.³⁷ Respondents were also invited to elaborate on their answers in writing, to provide further context to their responses. Recent (post-2015) literature on CDR was reviewed to identify 60 academics covering a range of topics, expertise, and

perspectives on CDR. Of these, 16 responded to the survey, providing a sample size comparable to other structured elicitations in the energy futures literature.^{38–41} The survey was conducted between 12th October 2020 and 2nd November 2020. The full survey text, list of respondents and further detail on the treatment of results are provided in [Notes S1](#) and [S2](#). [Table S1](#) provides the list of expert respondents, and [Table S2](#) explores some of the steps taken to ensure best practice in expert elicitation was followed during this exercise.

This analysis compares the survey results with publicly available model outputs ([Note S3](#) and [Table S4](#)), which may lag the best understanding of CDR potential in the IAM community. Work is underway to improve model representation of CDR, but challenges remain.⁴² Further analysis could compare the survey results to new IAM outputs, most notably the AR6 scenario database, when this becomes available.

TIAM-Grantham

The survey was used to produce a probability distribution for the level of CO₂ removal feasible via each method. This was then used to perform stochastic optimization in TIAM-Grantham,⁴³ a detailed-process IAM. TIAM-Grantham is a least-cost optimization model, which selects the portfolio of technologies that will meet future energy service demands at minimum cost, subject to user-defined constraints such as cumulative carbon budgets. A 2018–2100 carbon budget of 580GtCO₂ was applied, consistent with limiting end-of-century warming to 1.5°C with a 50% likelihood.² IAMs have historically underestimated the role of demand reduction in limiting warming.⁴⁴ TIAM-Grantham here incorporates a bottom-up assessment of the potential for demand reduction in key end-use sectors.⁴⁵ Three CDR options are represented in TIAM-Grantham: BECCS, DACCS, and AR. AR can be deployed from 2020, while BECCS and DACCS deployment starts in post-2030.

All scenarios use SSP2 drivers^{46,47} and include a limit on sustainable biomass of 100 EJ/year in 2050, growing to 130 EJ/year by 2100 to represent increased yields and reduced competition for land due to dietary shifts. This represents a biomass potential for which there is large agreement in the literature.⁴⁸ All scenarios account for the current climate and energy policies of the G20 nations,⁴⁹ and then transition to coordinated action to reduce emissions in 2020, with climate policy represented by a global carbon price.

TIAM-Grantham reports anthropogenic CO₂ emissions from fossil fuel and industry (FFI) and land-use, land-use change and forestry (LULUCF). To estimate corresponding trajectories for major non-CO₂ gases (CH₄ and N₂O), we use Silicone, an open-source package developed to help infer missing species data from scenarios.⁵⁰ We use an infiller database which consists of greenhouse gas trajectories for all scenarios which limit warming to 1.5°C with no or low overshoot in the SR1.5 database and use a quantile-rolling-windows cruncher to infill data for CH₄ and N₂O, although we also test the RMS closest cruncher. These are converted to CO₂eq using GWP₁₀₀ values from the IPCC's 5th Assessment report⁵¹ ([Note S4](#); [Table S5](#); [Figure S4](#)).

Stochastic analysis in TIAM-Grantham

Uncertainty in CDR potential is resolved post-2030. Using stochastic optimization, TIAM-Grantham calculates a hedging strategy for the 2020–2030 period which minimizes the expected cost over all possible future states-of-world (SOWs).⁵² For j future possible states of the world, the expected cost of strategy k is given by [Equation 1](#), where p_j is the probability associated with the j^{th} state of the world, and $\text{Cost}_{j|k}$ is the system cost of following strategy k and the j^{th} state of the world being realized.

Table 2. The relationship between the level of CDR uncertainty and the level of sampling from the expert-derived CDR probability distribution

Level of CDR uncertainty	Number of future CDR potentials considered
None	one (median of the distribution)
Low	two (25 th and 75 th percentiles of the distribution)
Medium	three (16 th , 50 th , and 83 rd percentiles of the distribution)
High	four (13 th , 38 th , 63 rd , and 88 th percentiles of the distribution)

$$\overline{C}_k = \sum_j p_j \times \text{Cost}_{jk} \quad (\text{Equation 1})$$

In this analysis, the j SOWs represent different possible future CDR potentials, and the k strategies represent different emission targets for 2030 (and associated action to achieve these targets). The more future SOWs included, the greater the representation of uncertainty in the scenario. We represent the survey data via a single SOW (deterministic scenario where CDR availability is known with certainty), two SOWs (high-/low- CDR availability), three SOWs (high-/medium-/low- CDR availability), and four SOWs (low-/medium-low-/medium-high-/high- CDR availability). Table 2 presents these four levels of CDR uncertainty, and what they mean in terms of sampling from the expert-derived probability distributions.

Further sampling of the distribution beyond the illustrative scenarios presented here would likely lead to even greater emission reductions, by better representing the tails of the distribution. Computational power limits this analysis to considering a max of four future CDR potentials simultaneously via stochastic optimization.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.joule.2021.09.004>.

ACKNOWLEDGMENTS

A.G., A.H., and S.M. acknowledge support from the H2020 European Commission Project PARIS REINFORCE (grant no. 820846). N.G. was supported by the Natural Environment Research Council (NERC) (grant no. NE/L002515/1) as well as by the Department for Business, Energy and Industrial Strategy (BEIS). We would like to thank Glen Peters, Joeri Rogelj, and Isabela Butnar for valuable feedback that helped shape the expert survey design.

AUTHOR CONTRIBUTIONS

N.G. conceived of the initial research design, performed the modeling with support from S.M., and led the paper writing. A.G. and A.H. provided guidance and input on the development of the research hypothesis and supported the expert survey and model analysis. All authors provided feedback and contributed to writing the paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: April 1, 2021

Revised: July 6, 2021

Accepted: September 8, 2021

Published: October 20, 2021

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